

A New Water Hammer Decay Model: Analyzing the Interference of Multiple Fractures and Perforations on Decay Rate

Yinghao Luo¹ , Xiaodong Hu^{1*} , Fujian Zhou¹ , Yang Qiu¹ , Xutao Lu¹ , Jinyuan Li¹ , and Yajing Wang¹ 

¹China University of Petroleum-Beijing

Summary

Hydraulic fracturing technology is an important technique for developing tight oil and gas resources. Multicluster fracture diagnosis is significant for hydraulic fracture design and optimization. Water hammer fracturing diagnostic technology is an emerging diagnostic technology. Fracture network complexity can be assessed by the water hammer decay rate. In this paper, a new characteristic line coupling model for the fracture-wellbore system is proposed. This model calculates the number of fractures and perforations based on the water hammer signal. First, our paper defines a method for calculating the water hammer decay rate. Second, a characteristic line simulation method for water hammer coupled with fractures and perforations is proposed. Third, the effects of different fracture-wellbore systems on the water hammer decay rate are analyzed. The method is validated by laboratory experiments and numerical analysis. Finally, through a field case, the results of the number of fracture openings explained by the optical fiber and our model are compared to provide support for the field water hammer fracturing diagnosis. Our paper provides a new model for water hammer decay rate calculation and evaluation of fracture number.

Introduction

At present, the proven tight oil and gas resources are abundant, and the development of tight oil and gas resources has gradually become the focus of oil and gas development (Wang et al. 2016). Tight oil and gas resources have low reservoir permeability, poor formation connectivity, and low recovery (Scanlon et al. 2014; Song et al. 2017). The rise of hydraulic fracturing technology has greatly contributed to the successful development of tight oil and gas resources (William et al. 2018; Dong et al. 2019; Sadykov et al. 2019; Guo et al. 2021; He et al. 2020). Hydraulic fracturing requires pumping a high-pressure fluid into the formation through perforations to develop hydraulic fractures. The fractures developed by hydraulic fracturing will greatly influence the oil and gas recovery rate (Ciezobka et al. 2016; Iriarte et al. 2017; Dung et al. 2021). Therefore, it is important to evaluate the fracturing effect and fracture network structure of hydraulic fracturing (Cipolla and Wright 2002).

However, because hydraulic fractures are underground, fracture diagnosis often requires additional equipment. The water hammer fracturing diagnostic technique collects the water hammer pressure signals during the pump shutdown stage. The water hammer propagates through the wellbore and interacts with the downhole fractures to affect the water hammer curve (Liang et al. 2017). The application of this technique has been verified by field analysis (Bogdan et al. 2016; Korkin et al. 2020). At the same time, the technology has high economic benefits, rapid analysis, and easy operation. These advantages enable the technology to be applied on a large scale (Parkhonyuk et al. 2018).

The idea of diagnosing bottomhole fractures with water hammer signals was first proposed by Holzhausen and Gooch (1985) through pressure wave excitation experiments at the wellhead. In subsequent field tests, the idea proved to be adaptable. In Holzhausen's model, the fracture is described as a series circuit of resistance R (representing the flow resistance pressure drop between the wellbore and fracture) and capacitance C (representing the fluid storage capacity of the fracture). Subsequently, Paige et al. (1995) added an inductance I to Holzhausen's model to represent fluid mass variation in fractures. Mondal (2010) proposed a model to represent bottomhole boundary conditions between wellbore and fracture (referred to here as the RCI model) by using circuit simulation of fracture conditions. The authors used method of characteristic combined with bottomhole boundary conditions (R , C , I) to numerically solve the wellbore water hammer. This method can be used to solve the average bottomhole net fracture pressure and fracture geometry by R , C , and I . On this basis, Carey et al. (2015) proposed that in the case of single-segment multicluster fracturing, the wellbore-fracture boundary condition proposed by Mondal is an equivalence of multiple fracture effects into a single fracture. In the following year, Carey found a strong correlation between the water hammer model and the fracture length interpreted by microseismic (Carey et al. 2016). In the same year, Ciezobka et al. (2016) presented an idea related to the decay rate of water hammer. The authors considered the energy absorbed by the fracture network when the pressure pulse passes through it. If the size of the fracture network is larger, the water hammer decay should also be higher. Also, the authors point out that additional fractures or a larger fracture network will make the water hammer decay faster. In this paper, Mondal and Carey models and Ciezobka's idea of water hammer decay rate are introduced to solve for the number of fracture openings. Thereafter, Hwang et al. (2017) investigated the effect of each parameter in the RCI model on the water hammer signal and applied the RCI model to multicluster fracturing. Haustveit et al. (2017) compared the results of electromagnetic imaging, microseismic, pressure disturbance testing, fracture simulation, and water hammer pressure diagnostics and noted that the fracture lengths estimated by water hammer model inversion were longer than those estimated by other methods. This may be because the water hammer pressure diagnosis results considered only one "effective" fracture (Haustveit et al. 2017). Dung et al. (2021) analyzed the effect of each construction parameter on the water hammer signal during hydraulic fracturing by numerical simulation methods. Hu et al. (2022) introduced stress interference between fractures under the wellbore-fracture boundary condition proposed by Mondal and proposed a water hammer diagnostic method capable of calculating multiple fracture dimensions. In the RCI model, the resistance R is introduced as an important

*Corresponding author; email: huxiaodong@cup.edu.cn

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parameter affecting water hammer decay for matching water hammer. However, the mainstream RCI models are more concerned with the volume of fracture modification. With the in-depth study of multifracture modifications, the disadvantage of resistance R is that it can only represent the effect of an equivalent single fracture. In the more complex multifracture case, the resistance R is no longer able to represent the impact of the number of fractures after the fracturing, but the number of fractures is correlated with the water hammer decay. Ciezobka and Iriarte proposed the assumption that the water hammer decay rate is correlated with the fracture network and that additional fracture expansion will lead to larger water hammer decay (Ciezobka et al. 2016; Iriarte et al. 2017). Dung summarized the water hammer and production data to conclude that wells with shorter water hammer durations are more productive. However, Dung's conclusion lacks a quantifiable indicator of water hammer decay rate and cannot be widely used in water hammer fracture diagnosis (Dung et al. 2021). The studies of the abovementioned scholars have sufficiently demonstrated the correlation between water hammer decay and fracture networks. Therefore, the decay rate study is in urgent need of a scientific method for calculating the decay rate and a method that can systematically analyze the relationship between fracture and water hammer decay rate.

The two main factors affecting the water hammer decay rate are the pipe flow resistance and the wellbore-fracture flow resistance. In the calculation of water hammer, pipe flow resistance has been studied by many scholars, and mature calculation methods have been proposed (Chaudhry 2014). However, in the wellbore-fracture model represented by RCI, the flow resistance between the wellbore and the fracture is represented only by the resistance R (Carey et al. 2015). The flow resistance between the wellbore and the fracture has a complex composition and lacks a clear physical meaning expressed only by the resistance R to determine the number of fractures. It is also difficult to properly represent the power relationship between pressure drop and flow rate by resistance R , which may lead to poor water hammer matching. In addition, the relationship between wellbore-fracture flow resistance and water hammer decay rate has not been analyzed, and the water hammer decay rate lacks a scientific calculation method.

Therefore, this paper proposes a new method to consider the wellbore-fracture flow resistance, which overcomes the problem of unclear physical meaning of the resistance R . The method expresses the pressure drop of the flow resistance between the wellbore and fracture due to the resistance R as the pressure drop brought by the fluid passing through the perforations and fractures (including the perforation friction pressure drop and the fracture tortuous friction pressure drop). Based on this, a water hammer decay rate calculation method is also proposed in this paper. Then, the water hammer curve is simulated based on the water hammer model proposed in this paper considering fractures and bottomhole perforations. An inverse fracture number calculation method is obtained by matching the field water hammer decay rate. Then, the model is validated by experimental and numerical simulation methods in this paper. Finally, the model is used in a multicluster hydraulic fracturing field. A comparative study between the model in this paper and the fiber-optic monitoring interpretation is carried out. The research in this paper can provide a diagnostic method for pump stoppage water hammer diagnosis method regarding water hammer decay rate and fracture number.

Model Development

The water hammer decay rate analysis model proposed in this paper will be divided into two parts. In the first part, we define the process and method of calculating the decay rate to provide a method for subsequent decay rate calculations. The method extracts the key points in the water hammer data and fits the data points after normalization to obtain the water hammer decay rate data. In the second part, we propose a method to define the boundary conditions of bottomhole fractures considering fractures and perforation friction. Water hammer is solved by the method of characteristic. Meanwhile, a laboratory experiment on the relationship between the number of fractures and the water hammer decay rate is designed.

Decay Rate Calculation Method. This part is based on the extraction, normalization, and curve fitting of water hammer data to obtain a water hammer decay rate parameter method. The water hammer signal will generate cycles. Each cycle will produce peaks and valleys. The method first extracts the peak of each cycle in the water hammer data and records the pressure P_i ($i = 1, 2, \dots, n$) and time T_i ($i = 1, 2, \dots, n$) of the peak of each cycle. The peak is defined as the maximum point in one cycle of the water hammer wave. Second, the recorded pressures and times need to be standardized. The time of the first peak point is used as the zero starting point of the time axis, and the time of each subsequent peak is redefined in this time axis (in seconds). The average pressure after the water hammer stops is defined as the water hammer stop pressure (P_{stop}), which is the minimum value normalized for other pressures. The calculation formula can be expressed as Eqs. 1 and 2.

$$T_i = T_i - T_1, \quad (1)$$

$$P_i = (P_i - P_{\text{stop}}) / (P_1 - P_{\text{stop}}), \quad (2)$$

where P_1 is the point in the cycle with the maximum peak pressure and T_1 is the time corresponding to the maximum pressure point P_1 . Iriarte et al. (2017) note that the water hammer signal decay can be represented by a curve fit from Eq. 3. Fig. 1 shows the selection of the peak pressure and the expression of the decay rate curve under the decay rate method in this paper. Among them, the analytical value of Eq. 3 has been normalized by Eqs. 1 and 2. The value of A is approximately equal to 1, and the value of B can indicate the decay rate of water hammer. The larger the value of B , the more obvious the water hammer decay is.

$$P_i = A \times e^{-T_i B}. \quad (3)$$

Water Hammer Model Considering Fractures and Perforations. The model is used to simulate the water hammer process of pump stoppage by the model of characteristic. The model consists of two main parts: One is the modeling of the flow in the wellbore and the other is the establishment of the boundary conditions. The wellbore flow model is mainly controlled by the momentum equation and the continuity equation. Chaudhry proposed that the momentum equation and the continuity equation can be simplified as Eqs. 4 and 5 in the case of low Mach number unsteady flow (Chaudhry 2014).

$$\frac{\partial Q}{\partial t} + gA \frac{\partial H}{\partial x} + \frac{fQ|Q|}{2DA} = 0, \quad (4)$$

$$\frac{\partial H}{\partial t} + \frac{a^2}{gA} \frac{\partial Q}{\partial x} = 0, \quad (5)$$

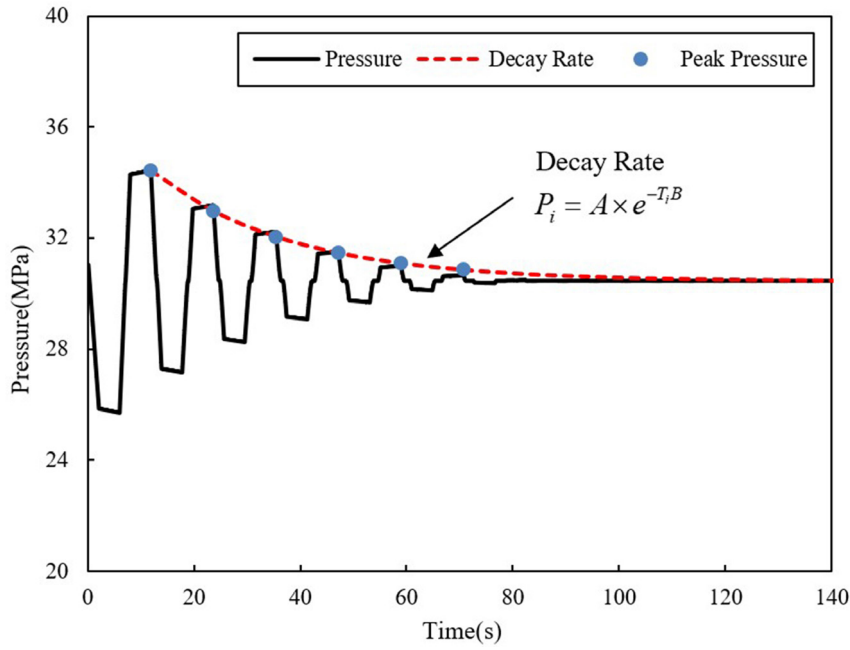


Fig. 1—Water hammer signal decay.

where in Q denotes fluid flow rate (m^3/s), H denotes the pressure head (m), t denotes the time (seconds), x denotes the coordinate axis along the axis of the pipe (m), D denotes the pipe diameter (m), g denotes the acceleration of gravity (m/s^2), f denotes the friction coefficient, A denotes the cross-sectional area of the wellbore (m^2), and a denotes the wave speed (m/s).

The momentum and continuity equations are transformed into four ordinary differential equations by the method of characteristic. Wylie and Streeter begin the method of characteristic transformation by rewriting the momentum and continuity equations, and the characteristic lines C^+ and C^- can be expressed by Eqs. 6 and 7 (Wylie and Streeter 1993).

$$C^+: H_i^{t+\Delta t} = H_{i-1}^t - B \left(Q_i^{t+\Delta t} - Q_{i-1}^t \right) - r Q_{i-1}^t |Q_{i-1}^t|, \quad (6)$$

$$C^-: H_i^{t+\Delta t} = H_{i+1}^t + B \left(Q_i^{t+\Delta t} - Q_{i+1}^t \right) + r Q_{i+1}^t |Q_{i+1}^t|, \quad (7)$$

where $B = \frac{a}{gA}$ and $r = \frac{f\Delta x}{2gDA^2}$. The description of the method of characteristic has been very well described in previous studies by scholars, and only the key formulas used for the solution calculation are enumerated in this paper (Wylie and Streeter 1993). However, Eqs. 6 and 7 are only valid when both C^+ and C^- equations are present. At the wellhead and bottom of the hole, only one of the C^+ or C^- equations are present, so boundary conditions are required for the solution.

The boundary conditions of the water hammer model containing fractures are divided into two main parts: the wellhead boundary conditions and the bottomhole fracture boundary conditions. At the wellhead, the flow at the wellhead is assumed to decrease linearly to zero during pump shutdown. Therefore, the boundary condition of the wellhead can be defined by the change in flow rate. According to the wellhead boundary flow rate, we can calculate the wellhead boundary water head through the C^- equation.

In a bottomhole fracture, the bottomhole fracture boundary condition can consist of a set resistance R , capacitance C , and inductance I equivalent to a circuit (RCI model), according to the description of the bottomhole fracture boundary in Carey et al. (2015). To simplify the bottomhole pressure inversion process, we make the following assumptions: (1) Only plane fractures of constant height and ellipsoidal section are created; (2) the fracture length is constant, and the fracture compliance during the water hammer duration depends only on the change of width; (3) the leakoff of cracks is not considered during the water hammer duration; (4) the flow resistance mainly comes from the wellbore and near-wellbore areas, including wellbore flow, near-wellbore tortuosity, and perforation friction, without considering the flow resistance along the fracture length; (5) the average bottomhole pressure (P_{BH}) can be estimated by a water hammer model, ground pressure, or downhole pressure gauge (Carey et al. 2015).

According to the RCI model, the bottomhole pressure (P_{BH}) in the bottomhole boundary condition can be expressed by Eq. 8 (Carey et al. 2015).

$$P_{BH} = \Delta P_{nwf} + \Delta \bar{P}_0 + S_{H_{\min}} = \Delta P_{nwf} + \bar{P}_f. \quad (8)$$

P_{BH} denotes the bottomhole pressure, ΔP_{nwf} denotes the near-well friction pressure drop, $\Delta \bar{P}_0$ denotes the average net pressure, $S_{H_{\min}}$ denotes the minimum formation stress, and $\bar{P}_f$ denotes the average fracture pressure. ΔP_{nwf} can be expressed by the resistance R through Eq. 9 (Carey et al. 2015). The resistance R is defined as a constant of proportionality between the discharge Q into or out of the fracture and the potential difference required to maintain the discharge. $\Delta \bar{P}_0$ can be calculated by capacitance C and inductance I in the RCI model according to different fracture types (Shlyapobersky et al. 1988; Carey et al. 2015).

$$\Delta P_{nwf} = RQ_{\text{wellbottom}}. \quad (9)$$

where $Q_{\text{wellbottom}}$ denotes the flow rate at the bottom of the hole. It should be noted that the calculation method of bottomhole fracture boundary conditions of RCI model is described in detail in many papers (Carey et al. 2015, Carey et al. 2016; Hwang et al. 2017; Hu et al. 2022). This paper is to study the influence of near-well friction pressure drop (ΔP_{nwf}) on water hammer decay rate. Therefore, this paper only shows the calculation equation of near-well friction pressure drop (ΔP_{nwf}) in the RCI model, and the calculation method of the average net pressure ($\Delta \bar{P}_0$) can be found in Hu et al. (2022).

According to Ciezobka et al. (2016), additional fractures, more extensive fracture networks, and well-fracture connectivity can have an impact on the water hammer decay rate, and more complex fracture network conditions may lead to increased water hammer decay rates. Therefore, the near-well frictional pressure drop (ΔP_{nwf}) can have an impact on water hammer decay. The near-well frictional pressure drop is mainly caused by near-wellbore tortuosity and perforation friction resistance (Cleary et al. 1991). Therefore, in this paper, to investigate the specific law of near-well frictional pressure drop on water hammer decay, the pressure drop due to near-wellbore tortuosity (ΔP_{frac}) and perforation friction resistance (ΔP_{perf}) are applied to the bottomhole fracture boundary conditions for the first time. The near-wellbore tortuosity and the perforation friction pressure drop can be expressed by Q^α (Willingham et al. 1993; Romero et al. 1995):

$$\Delta P_{\text{frac}} = K_{\text{frac}} Q^{0.5}, \quad (10)$$

$$\Delta P_{\text{perf}} = K_{\text{perf}} Q^2, \quad (11)$$

where Q denotes the total flow rate at the bottom of the well (m^3/s), K_{frac} denotes the near-wellbore tortuosity coefficient, and K_{perf} denotes the perforation friction coefficient. The perforation friction coefficient can be expressed by Eq. 12 (Willingham et al. 1993):

$$K_{\text{perf}} = 0.807249 \frac{\rho}{n_p^2 D_p^4 C_d^2}, \quad (12)$$

where n_p denotes the number of perforations, D_p denotes the perforation diameter (m), C_d denotes the perforations flow coefficient, and ρ denotes the fluid density (kg/m^3). In this paper, Eq. 10 can be rewritten to represent the near-wellbore tortuosity pressure drop caused by different number of fractures, assuming that the number of fracture openings is i and each cluster of fractures is uniformly fluid:

$$\Delta P_{\text{frac}} = i \times \left(K_{\text{frac}} (Q/i)^{0.5} \right). \quad (13)$$

Therefore, according to Eqs. 8, 11, and 13, the bottomhole pressure can be expressed as

$$P_{\text{BH}} = \Delta \bar{P}_0 + S_{H_{\text{min}}} + \Delta P_{\text{nwf}} = \bar{P}_f + \Delta P_{\text{nwf}}, \quad (14)$$

$$P_{\text{BH}} = \bar{P}_f + 0.807249 \frac{\rho}{n_p^2 D_p^4 C_d^2} Q^2 + i \times \left(K_{\text{frac}} (Q/i)^{0.5} \right), \quad (15)$$

where $\Delta \bar{P}_0$ can be calculated by the RCI model described above, $\bar{P}_f = \Delta \bar{P}_0 + S_{H_{\text{min}}}$, and ΔP_{nwf} is replaced by $\Delta P_{\text{frac}} + \Delta P_{\text{perf}}$. Because the bottomhole pressure P_{BH} is a function related to Q , the bottomhole fracture boundary condition should be solved for flow rate by coupling Eq. 15 with Eq. 6, as shown in Eq. 16. Finally, the solved bottomhole fracture boundary flow rate is brought into Eq. 6 to solve for the bottomhole fracture boundary head.

$$\begin{aligned} & \left(\bar{P}_f + 0.807249 \frac{\rho}{n_p^2 D_p^4 C_d^2} (Q_i^{t+\Delta t})^2 + i \times \left(K_{\text{frac}} (Q_i^{t+\Delta t}/i)^{0.5} \right) \right) / (\rho g) \\ & = H_{i-1}^t - B \left(Q_i^{t+\Delta t} - Q_{i-1}^t \right) - r Q_{i-1}^t |Q_{i-1}^t| \end{aligned} \quad (16)$$

Based on the boundary conditions described above, a water hammer model that can consider fractures and perforations at the bottom of the hole can be obtained by using the method of characteristic. The different water hammer simulation curves can be obtained for different numbers of fractures and perforations. The water hammer decay rate can be calculated using the decay rate calculation method for different numbers of fractures and perforations.

Water Hammer Experiments

In this paper, a laboratory experimental system for simulating fracturing stop pump water hammer was built. The water hammer pressure wave decay rate test in the simulated horizontal pipe section was carried out. The water hammer decay variation law under different simulated fracture opening cases was studied. The device is mainly composed of a piping system, a control system, a collection system, and a fracture simulation system, and its working schematic diagram is shown in Fig. 2.

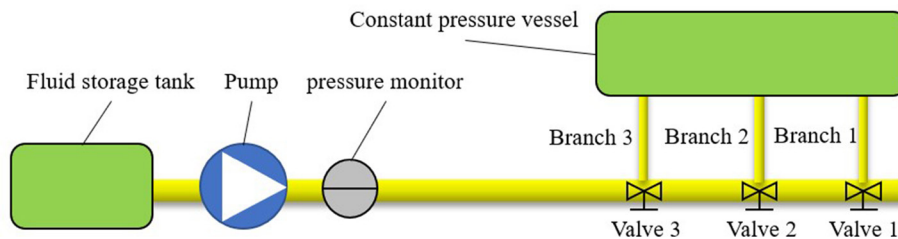


Fig. 2—Schematic diagram of the experimental device for simulating fractures.

Experimental Setup. The piping system is a 190-m-long welded cast iron pipeline with an inner diameter of 16 mm and a wall thickness of 1 mm. The control system is a screw pump, through which the liquid is pumped into the piping system, and the instantaneous stopping of the screw pump simulates the water hammer effect at the end of hydraulic fracturing. The collection system is a pressure monitor connected to the outlet of the screw pump at 2 m and the collection software. The range of the pressure monitor is 0 to 5 MPa, with an accuracy of 0.01% of full scale and a frequency response of 10 kHz. The fracture simulation system consists of three valve-connected stubs. The end of the branch pipe is connected to a 5-m-high constant pressure vessel to simulate the fracture pressure. The branch pipes are 15.9, 33.2, and 53.6 m from the end of the piping system. In this experiment, the water hammer decay of different branch pipes with the same flow rate was recorded to analyze the water hammer decay law.

Procedures. The laboratory experiment is to test the influence of fractures on water hammer decay rate by opening different numbers of simulated fractures. The first step is to connect the pressure monitor to the pipeline to ensure that the pump stop water hammer can be successfully collected. The fluid storage tank is filled with fluid to ensure that the experiment can continue. The second step is to open the corresponding branch valve according to the experimental scheme. There are seven groups of experimental schemes, which are opening Valves 1, 2, 3, Valves 1 and 2, Valves 1 and 3, Valves 2 and 3, Valve 1 only, Valve 2 only, and Valve 3 only, respectively. The third step is to open the screw pump with a working frequency of 20 Hz, waiting for stable flow in the pipeline. The fourth step is to use the pump operation interface to stop the pump instantaneously and use the high-frequency pressure monitor to collect the pressure data in the pipeline. Finally, repeat the above steps until all experimental measurements are completed. The experiment data interpretation and results will be discussed in the Model Results section.

Model Validation

To verify the feasibility of the model proposed in this paper in terms of water hammer simulation calculation and water hammer decay rate calculation, we cite the results of Carey's RCI model matching water hammer. The water hammer decay rate is also calculated to analyze whether the decay rate calculation method is accurate.

The RCI parameters for matching Well A in the Carey paper are shown in **Table 1** (Carey et al. 2015). **Table 1** also shows the key parameters used in this paper to replace the resistance R . The values of water hammer decay rate for the field data, RCI model, and the model in this paper can be seen in **Table 2**, and the corresponding water hammer curve comparison can be seen in **Fig. 3**.

Parameters	RCI Model Values	Our Model Values
Resistance, R (Pa·s/m ³)	1.52×10^7	—
Perforation flow coefficient	—	0.6
Perforation diameter (m)	—	0.01
Number of perforations	—	30
Tortuosity coefficient	—	550,000
Number of fractures	—	3
Near-wellbore friction, ΔP_{nwf} (MPa)	1.94	4.44
Capacitance, C (m ³ /Pa)	1.12×10^{-5}	1.12×10^{-5}
Inductance, I [Pa/(m ³ /s ²)]	1.75	1.75

Table 1—Parameters of wellbore-fracture boundary conditions.

Data	Field Data	Simulated Data by Carey	Simulated Data by Our Model
Decay rate	0.0523	0.0493	0.0507

Table 2—Decay rate for model validation.

Through the above case, it can be verified that the model proposed in this paper can be applied to simulate water hammer. Similar water hammer curves can be obtained when using the wellbore-fracture boundary conditions defined in this paper. The overall average pressure of the field data, RCI simulation, and the model in this paper are 29.5, 29.1, and 29.3 MPa, and the overall pressure of the model in this paper is 0.68% compared with the field data; the average pressures of the first five wave peaks of the above model are 31.8, 31.3, and 31.6 MPa, and the average pressure of the wave peaks in this paper is 0.63% compared with the field data. The initial wellhead pressures for the above models were 37.3, 34.5, and 37.0 MPa. The initial wellhead pressure RCI model was compared to the field data with an error of 7.5%. This may be because the near-wellbore friction resistance calculated by the resistance R is too small during the initial pumping stop stage. The difference between the values of near-wellbore friction pressure drop in the initial stage of this paper model and the RCI model is 2.5 MPa, which may be the reason why the initial pressure of the RCI model is lower than the field data. In addition, the water hammer decay rates of the above models are 0.0523, 0.0493, and 0.0507, and the decay rate error between this model and the field data is 0.0016, which is less than the decay rate error of the RCI model of 0.003. The model in this paper has a small error in calculating the water hammer pressure but is more accurate in calculating the initial wellhead pressure and matching the decay rate, so this case can confirm the feasibility of the method and model used in this paper.

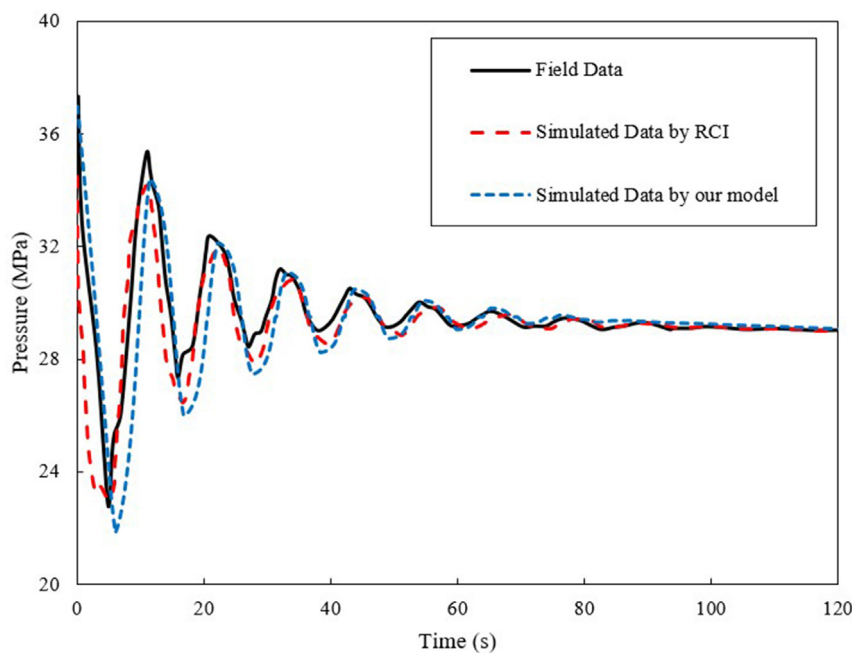


Fig. 3—Model validation of water hammer curve.

Model Results

In this paper, we first analyze the influence of the number of fractures on the water hammer decay rate by the model and then use experimental comparison to examine the effect law. Second, the water hammer curves are simulated for different bottomhole fractures and perforation parameters. Based on the simulated water hammer decay rate, radar plots are drawn to analyze the influence rule and degree of each parameter on water hammer decay. Finally, this paper combines the fiber-optic interpretation. The model decay rate, the number of fracture bars, and the fiber-optic interpretation are compared in this paper.

Base Case. Simulation Analysis. First, we created a simulated horizontal well with initial data for hydraulic fracturing. The initial data is given in Table 3. By changing the values of the number of fractures in the initial parameters, the water hammer decay rates taken for the cases of 1, 3, 5, and 7 fractures are obtained as represented in Fig. 4 and Table 4, and the corresponding water hammer curves are shown in Fig. 5.

Experimental Analysis. Through the water hammer decay rate laboratory experiments described above, the branches are named 1, 2, and 3 as shown in Fig. 2. By adjusting the number of branches open to simulate the number of fractures, Case 1 is open only for Branch

Parameters	Values
Pipe length (m)	4000
Vertical depth (m)	2500
Wave speed (m/s)	1000
Density (kg/m ³)	1000
Pipe diameter (m)	0.11
Initial flow rate (m ³ /min)	4
Pump stop time (seconds)	2
Flow friction	0.005
Perforation flow coefficient	0.6
Perforation diameter (m)	0.01
Number of perforations	25
Tortuosity coefficient	1,400,000
Number of fractures	3
Average fracture pressure (MPa)	60
Capacitance, C (m ³ /Pa)	1.5×10^{-5}
Inductance, I [Pa/(m ³ /s ²)]	2

Table 3—Simulated horizontal well water hammer initial parameters.

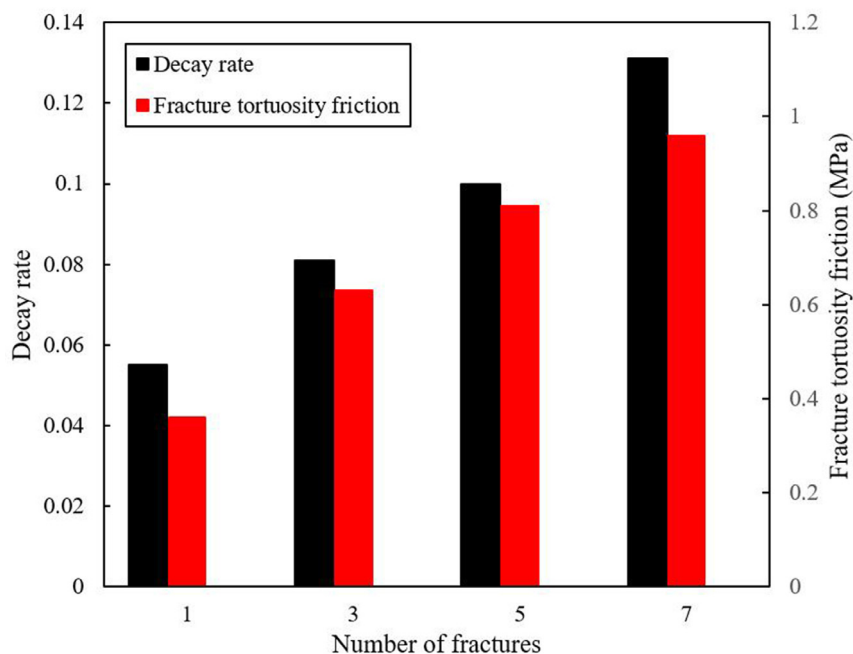


Fig. 4—Effect of the number of fractures on the decay rate and fracture tortuous friction.

Number of fractures	1	3	5	7
Decay rate	0.055	0.081	0.100	0.131

Table 4—Simulated fractures and decay rate results.

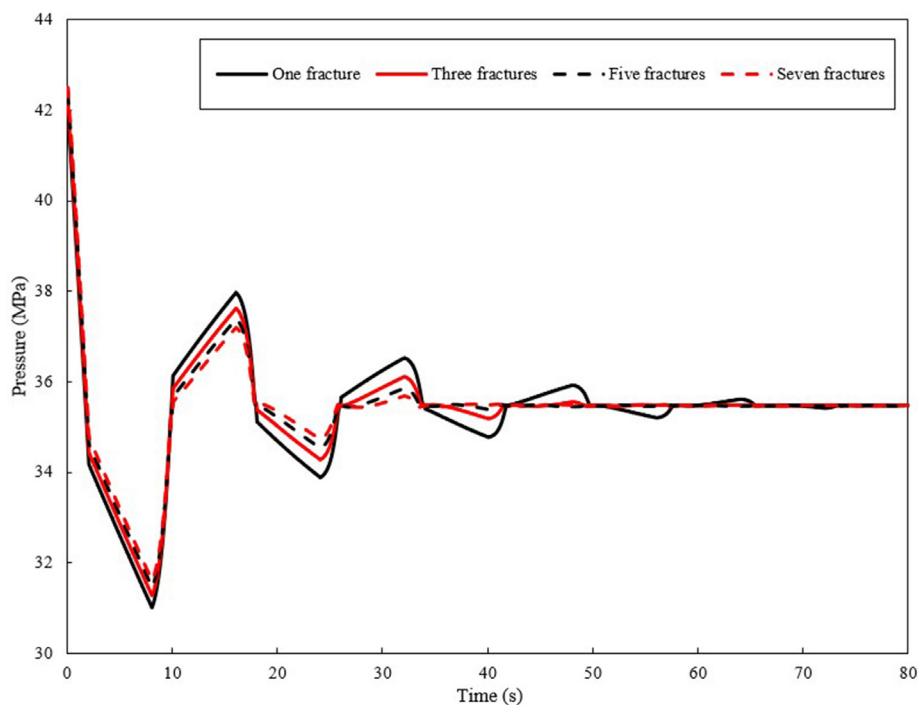


Fig. 5—Water hammer curves for different numbers of fractures.

1; Case 2 is open only for Branch 2; Case 3 is open only for Branch 3; Case 4 is open for Branches 1 and 2; Case 5 is open for Branches 2 and 3; Case 6 is open for Branches 1 and 3; Case 7 is open for Branches 1, 2, and 3. The experimental opening scheme of each case and the corresponding water hammer decay rate are shown in Table 5 and Fig. 6. The corresponding water hammer curves for each experimental case are shown in Fig. 7.

Cases	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7
Fracture opening	1	2	3	1, 2	2, 3	1, 3	1, 2, 3
Decay rate	0.350	0.288	0.209	0.528	0.398	0.473	0.643

Table 5—Experimental case parameters and decay rate results.

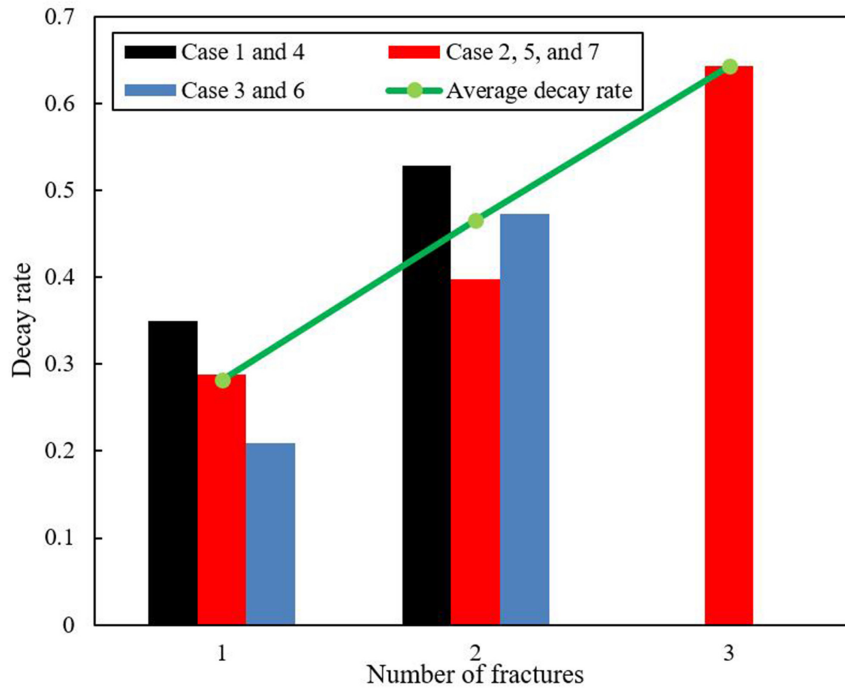


Fig. 6—Experimental case number of branches and decay rate results.

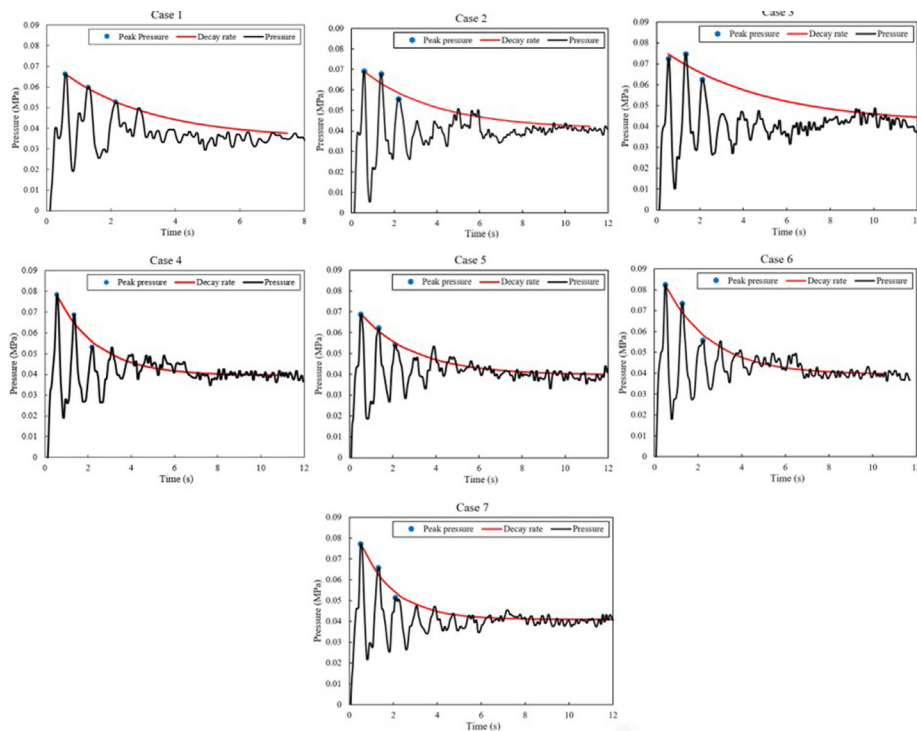


Fig. 7—Experimental case water hammer signal and decay rate fitting.

Based on the above base case analysis, it can be concluded that in the simulated case, when the number of fractures increases from 1 to 7, the water hammer decay rate is 0.055, 0.081, 0.100, and 0.131, and the fracture torsional friction pressure drop at the initial flow rate is 0.36, 0.63, 0.81, and 0.96 MPa. The increase of fracture torsional friction pressure drop leads to an increase in the water hammer decay rate. In the experimental case, the average water hammer decay rate is 0.282 when the number of branch pipe openings is 1, 0.466 when the number of branch pipe openings is 2, and 0.643 when the number of branch pipe openings is 3. Therefore, according to the simulation and experiment of this paper, the following conclusions can be made: (1) With the increase in the number of fractures, the water hammer decay rate will also increase; (2) the experimental water hammer decay rate increases with the increase in the number of branches, which is consistent with the law of simulation conclusions; and (3) the larger the experimental water hammer decay rate of a single branch, the larger the overall decay rate of the multibranches combination. For example, the decay rate on Branch 1 > Branch 2 > Branch 3, and the decay rate on Branches 1, 2, 3 > Branches 1, 2 > Branches 1, 3 > Branches 2, 3.

Numerical Case Analysis. In the initial data of Table 3, the parameters of the wellbore-fracture boundary conditions, including perforation diameter, number of perforations, perforation flow coefficient, tortuosity coefficient, number of fractures, average fracture pressure, inductance I , and capacitance C , were increased by 20% in this paper to compare the degree of influence of each parameter on water hammer decay rate. The effect law of each parameter on water hammer decay is shown in Fig. 8.

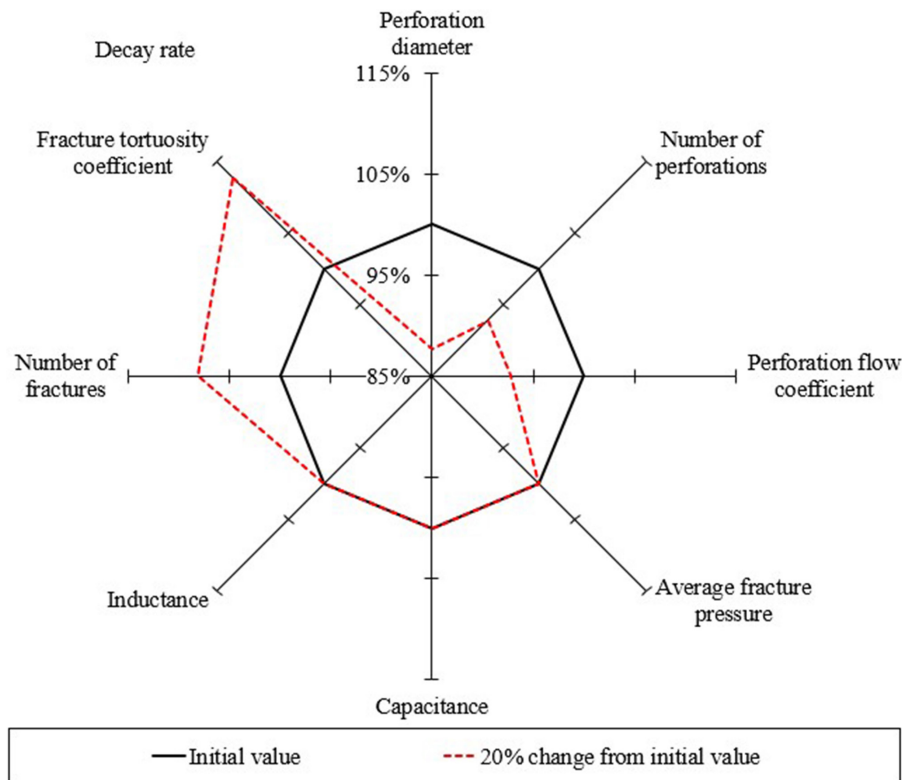


Fig. 8—Effect of bottomhole fracture parameters on decay rate.

Among them, the initial parameter decay rate is 0.081, and after increasing the parameters by 20%, the decay rate is 0.071 when the variable is perforation diameter, the number of perforations decay rate is 0.075, the perforation flow coefficient decay rate is 0.075, the tortuosity coefficient decay rate is 0.091, the number of fracture bars decay rate is 0.087, the average fracture pressure decay rate is 0.081, and the inductance decay rate is 0.81. According to the effect of parameters on the water hammer decay rate in Fig. 7, the following conclusions can be made: (1) Tortuosity coefficient and number of fractures are positively related to the decay rate. The decay rate increases with the increase of the tortuosity coefficient and the number of fractures. (2) The perforation diameter, number of perforations, and perforation flow coefficient are negatively correlated with the decay rate. The decay rate decreases with the increase of perforation diameter, number of perforations, and perforation flow coefficient. (3) Capacitance, inductance, and average fracture pressure are not correlated with the decay rate. Changes in capacitance, inductance, and average fracture pressure only affect the overall pressure. (4) Under the initial conditions of this paper, the tortuosity coefficient has the largest effect among the positively correlated parameters, with +12.8% change in decay rate. The perforation diameter has the largest effect among the negatively correlated parameters, with -12.3% change in decay rate.

Field Case Study

In this paper, the model is used to match the decay rate of the field data of a multifracture hydraulic fracturing horizontal well. The well is also equipped with optical fiber monitoring technology, which can monitor the bottomhole fractures in real time. The well is 3843 m deep. The well has different perforation schemes of 4 clusters, 10 holes per cluster and 6 clusters, 8 holes per cluster. In this paper, two fracturing stages of different perforation schemes are selected for analysis. The first stage is 4 clusters, and the second stage is 6 clusters. The value of each simulation parameter is determined by the matching results of the simulation and the field water hammer curve. Water hammer curve matching includes water hammer decay rate, amplitude, pump stop time pressure, pressure peak, waveform, and cycle matching. The water hammer decay rate is the most important determinant in the process of water hammer curve matching. The specific

simulation parameters and decay rate calculation results of each stage can be seen in **Table 6**. The comparison results of simulated water hammer and field water hammer in the corresponding stage can be seen in **Figs. 9 and 10**, and the optical fiber interpretation results can be seen in **Figs. 11 and 12**.

Parameters	Stage 1	Stage 2
Horizontal length (m)	912	484
Wave speed (m/s)	1470	1470
Pipe diameter (m)	0.11552	0.11552
Initial flow rate (m ³ /min)	3.7	2.2
Pump stop time (seconds)	3	2
Flow friction	0.0017	0.0015
Perforation flow coefficient	0.6	0.6
Perforation diameter (m)	0.01	0.01
Number of perforations	15	21
Tortuosity coefficient	1,650,000	1,700,000
Number of fractures	3	5
Average fracture pressure (MPa)	65.40	68.15
Simulated water hammer decay rate	0.0998	0.1195
Field water hammer decay rate	0.0996	0.1196

Table 6—Water hammer simulation parameters and decay rate results.

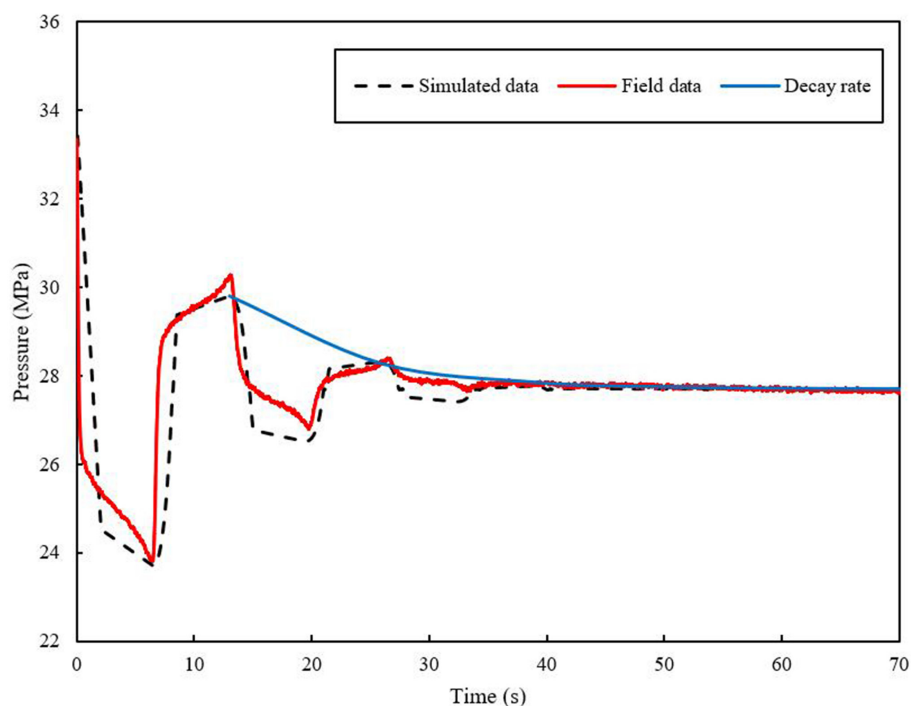


Fig. 9—Water hammer matching results and decay rates of Stage 1.

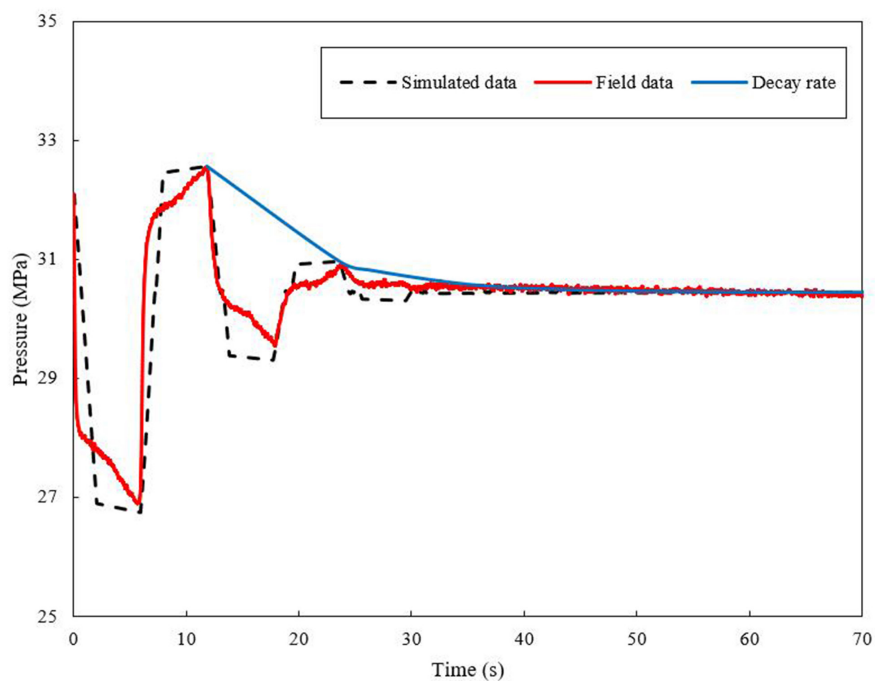


Fig. 10—Water hammer matching results and decay rates of Stage 2.

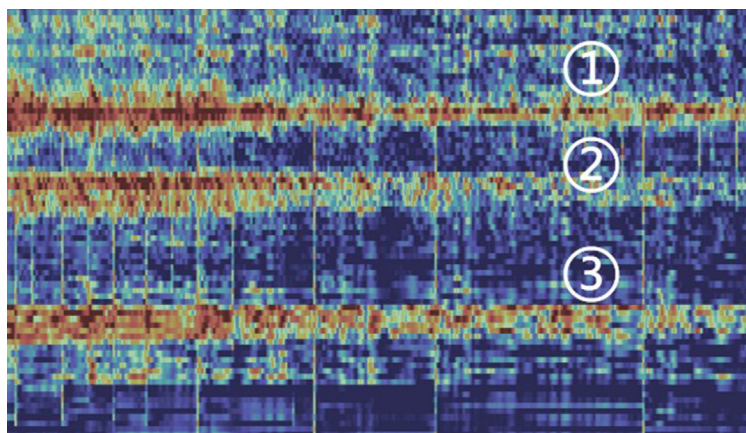


Fig. 11—Optical fiber signal diagram of Stage 1.

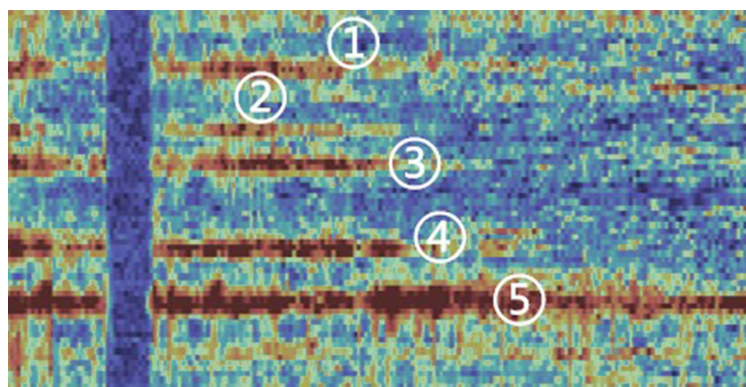


Fig. 12—Optical fiber signal diagram of Stage 2.

In the first stage, the decay rate of water hammer is 0.0996, the decay rate of simulated water hammer is 0.0998, and the decay rate error is 0.2%. In the second stage, the decay rate of water hammer is 0.1196, the decay rate of simulated water hammer is 0.1195, and the decay rate error is 0.08%. By matching the decay rate of water hammer, a set of bottomhole fracture and perforation parameters matching the decay rate of water hammer can be obtained. When the simulated water hammer decay is the same as or similar to the field water hammer decay, it can be considered that the wellbore-fracture boundary conditions under the simulation can approximately express the real bottomhole conditions. Through the analysis of the above two field cases and the matching results of fracture number and decay rate in **Table 6**, it can be concluded that there are three clusters of fractures in the first stage of fracturing and five clusters of fractures in the second stage of fracturing. According to the optical fiber monitoring liquid position, it can be observed that the first stage has three liquid positions; there are five liquid inlet positions in the second stage, which is related to the number of fractures calculated by the model. According to the field case study of the above water hammer decay rate, it can be concluded that the water hammer decay rate is related to the fracture reconstruction volume and the number of fractures. Therefore, a larger decay rate indicates a larger near-well friction, a larger decay rate may indicate more fractures.

Discussion

Through the research in this paper, it is found that there is a strong correlation between the water hammer decay rate and the number of fractures during the pump stopping process. The larger water hammer decay rate in the fracturing pump shutdown may represent the formation of more fractures during fracturing. At the same time, this paper proposes a new model. First, decay rate calculation model can correctly calculate the value of the water hammer decay rate from water hammer data. Second, our model considers the influence of near-wellbore tortuosity and perforation friction pressure drop in the bottomhole boundary condition obtained by the method of characteristic, so that the characteristic line method can simulate the water hammer of hydraulic fracturing with fractures and perforations. Compared with previous studies, we focus on the influence of flow friction near the well on water hammer decay rate. This paper presents a standard calculation method for the water hammer decay rate. At the same time, this paper considers the influence of fracture and perforation on the bottomhole boundary conditions, so that the method of judging the number of fractures by inverting water hammer decay can be realized. At present, this method can realize rapid fracturing diagnosis by analyzing the decay rate of water hammer and determine whether an effective fracture or fracture network structure is formed. However, the model in this paper only considers the flow friction near the well and does not consider the flow friction caused by the fracture geometry. Subsequent studies will consider the influence of fracture geometry on water hammer decay rate.

Conclusions

1. The increase in the number or complexity of fractures will result in an increase of the water hammer decay rate. In the same case, more fractures may show faster water hammer decay.
2. The increase of perforation number or perforation diameter will result in a decrease of water hammer decay rate. In the same case, more perforations may show slower water hammer decay.
3. The average fracture pressure will not affect the water hammer decay rate. The average fracture pressure needs to be judged with RCI model.
4. The effect of fracture on water hammer decay is positive, and the effect of perforation on water hammer is negative. Therefore, in the process of inversion of water hammer decay, the requirements of fracturing perforation parameters should be fully considered, and the numerical relationship between fracture and perforation should be reasonably selected for simulation.

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References

- Bogdan, A. V., Keilers, A., Oussoltsev, D. et al. 2016. Real-Time Interpretation of Leak Isolation with Degradable Diverter Using High Frequency Pressure Monitoring. Paper presented at the SPE Asia Pacific Oil & Gas Conference and Exhibition, Perth, Australia, 25–27 October. SPE-182451-MS. <https://doi.org/10.2118/182451-MS>.
- Carey, M. A., Mondal, S., and Sharma, M. M. 2015. Analysis of Water Hammer Signatures for Fracture Diagnostics. Paper presented at the SPE Annual Technical Conference and Exhibition, Houston, Texas, USA, 28–30 September. SPE-174866-MS. <https://doi.org/10.2118/174866-MS>.
- Carey, M. A., Mondal, S., Sharma, M. M. et al. 2016. Correlating Water Hammer Signatures with Production Log and Microseismic Data in Fractured Horizontal Wells. Paper presented at the SPE Hydraulic Fracturing Technology Conference, The Woodlands, Texas, USA, 9–11 February. SPE-179108-MS. <https://doi.org/10.2118/179108-MS>.
- Chaudhry, M. H. 2014. *Applied Hydraulic Transients*. New York, USA: Springer. <https://doi.org/10.1007/978-1-4614-8538-4>.
- Ciezobka, J., Maity, D., and Salehi, I. 2016. Variable Pump Rate Fracturing Leads to Improved Production in the Marcellus Shale. Paper presented at the SPE Hydraulic Fracturing Technology Conference, The Woodlands, Texas, USA, 9–11 February. SPE-179107-MS. <https://doi.org/10.2118/179107-MS>.
- Cipolla, C. L. and Wright, C. A. 2002. Diagnostic Techniques To Understand Hydraulic Fracturing: What? Why? And How? *SPE Prod & Fac* **17** (1): 23–35. SPE-75359-PA. <https://doi.org/10.2118/75359-PA>.
- Cleary, M. P., Wright, C. A., and Wright, T. B. 1991. Experimental and Modeling Evidence for Major Changes in Hydraulic Fracturing Design and Field Procedures. Paper presented at the SPE Gas Technology Symposium, Houston, Texas, USA, 22–24 January. SPE-21494-MS. <https://doi.org/10.2118/21494-MS>.
- Dong, J., Chen, M., Jin, Y. et al. 2019. Study on Micro-Scale Properties of Cohesive Zone in Shale. *Int J Solids Struct* **163**: 178–193. <https://doi.org/10.1016/j.ijsolstr.2019.01.004>.
- Dung, N., David, C., Tom, D. et al. 2021. Practical Applications of Water Hammer Analysis from Hydraulic Fracturing Treatments. Paper presented at the SPE Hydraulic Fracturing Technology Conference and Exhibition, Virtual, 4–6 May. SPE-204154-MS. <https://doi.org/10.2118/204154-MS>.
- Guo, T., Tang, S., Liu, S. et al. 2021. Physical Simulation of Hydraulic Fracturing of Large-Sized Tight Sandstone Outcrops. *SPE J* **26** (1): 372–393. SPE-204210-PA. <https://doi.org/10.2118/204210-PA>.
- Haustveit, K., Dahlgren, K., Greenwood, H. et al. 2017. New Age Fracture Mapping Diagnostic Tools-A STACK Case Study. Paper presented at the SPE Hydraulic Fracturing Technology Conference and Exhibition, Texas, USA, 24–26 January. SPE-184862-MS. <https://doi.org/10.2118/184862-MS>.

- He, Y., Cheng, S., Sun, Z. et al. 2020. Improving Oil Recovery Through Fracture Injection and Production of Multiple Fractured Horizontal Wells. *J Energy Resour Technol* **142** (5): 5. <https://doi.org/10.1115/1.4045957>.
- Holzhausen, G. R. and Gooch, R. P. 1985. Impedance of Hydraulic Fractures: Its Measurement and Use for Estimating Fracture Closure Pressure and Dimensions. Paper presented at the SPE/DOE Low Permeability Gas Reservoirs Symposium, Denver, Colorado, USA, 19–22 May. SPE-13892-MS. <https://doi.org/10.2118/13892-MS>.
- Hu, X., Luo, Y., Zhou, F. et al. 2022. Evaluation of Multi-Fractures Geometry Based on Water Hammer Signals: A New Comprehensive Model and Field Application. *J Hydrol* **612**: 128240. <https://doi.org/10.1016/j.jhydrol.2022.128240>.
- Hwang, J., Szabian, M. J., and Sharma, M. 2017. Hydraulic Fracture Diagnostics and Stress Interference Analysis by Water Hammer Signatures in Multi-Stage Pumping Data. Paper presented at the SPE/AAPG/SEG Unconventional Resources Technology Conference, Austin, Texas, USA, 24–26 July. URTEC-2687423-MS. <https://doi.org/10.15530/urtec-2017-2687423>.
- Iriarte, J., Merritt, J., and Kreyche, B. 2017. Using Water Hammer Characteristics as a Fracture Treatment Diagnostic. Paper presented at the SPE Oklahoma City Oil and Gas Symposium, Oklahoma, USA, 27–31 March. SPE-185087-MS. <https://doi.org/10.2118/185087-MS>.
- Korkin, R., Parkhonyuk, S., Fedorov, A. et al. 2020. High Frequency Pressure Monitoring and Data Analytics for Stimulation Efficiency Determination: New Perspectives or Potential Limits. Paper presented at the SPE Hydraulic Fracturing Technology Conference and Exhibition, The Woodlands, Texas, USA, 4–6 February. SPE-199762-MS. <https://doi.org/10.2118/199762-MS>.
- Liang, C., O'Reilly, O., Dunham, E. M. et al. 2017. Hydraulic Fracture Diagnostics from Krauklis-Wave Resonance and Tube-Wave Reflections. *Geophysics* **82** (3): D171–D186. <https://doi.org/10.1190/geo2016-0480.1>.
- Mondal, S. 2010. *Pressure Transients in Wellbores: Water Hammer Effects and Implications for Fracture Diagnostics*. MSc thesis, The University of Texas at Austin, Austin, Texas, USA.
- Paige, R. W., Murray, L. R., and Roberts, J. D. M. 1995. Field Application of Hydraulic Impedance Testing for Fracture Measurement. *SPE Prod & Fac* **10** (1): 7–12. SPE-26525-PA. <https://doi.org/10.2118/26525-PA>.
- Parkhonyuk, S., Fedorov, A., Kabannik, A. et al. 2018. Measurements While Fracturing: Nonintrusive Method of Hydraulic Fracturing Monitoring. Paper presented at the SPE Hydraulic Fracturing Technology Conference and Exhibition, The Woodlands, Texas, USA, 23–25 January. SPE-189886-MS. <https://doi.org/10.2118/189886-MS>.
- Romero, J., Mack, M. G., and Elbel, J. L. 1995. Theoretical Model and Numerical Investigation of Near-Wellbore Effects in Hydraulic Fracturing. Paper presented at the SPE Annual Technical Conference and Exhibition, Dallas, Texas, USA, 22–25 October. SPE-30506-MS. <https://doi.org/10.2118/30506-MS>.
- Sadykov, A., Baki, S., Mechkak, K. et al. 2019. Diversion Techniques Applications in Unconventional Resources Fields. Paper presented at the SPE Middle East Oil and Gas Show and Conference, Manama, Bahrain, 18–21 March. SPE-194791-MS. <https://doi.org/10.2118/194791-MS>.
- Scanlon, B. R., Reedy, R. C., and Nicot, J. P. 2014. Comparison of Water Use for Hydraulic Fracturing for Unconventional Oil and Gas versus Conventional Oil. *Environ Sci Technol* **48** (20): 12386–12393. <https://doi.org/10.1021/es502506v>.
- Shlyapobersky, J., Wong, G. K., and Walhaug, W. W. 1988. Overpressure Calibrated Design of Hydraulic Fracture Stimulations. Paper presented at the SPE Annual Technical Conference and Exhibition, Houston, Texas, USA, 2–5 October. SPE-18194-MS. <https://doi.org/10.2118/18194-MS>.
- Song, Y., Li, Z., Jiang, Z. et al. 2017. Progress and Development Trend of Unconventional Oil and Gas Geological Research. *Pet Explor Dev* **44** (4): 675–685. [https://doi.org/10.1016/S1876-3804\(17\)30077-0](https://doi.org/10.1016/S1876-3804(17)30077-0).
- Wang, H., Ma, F., Tong, X. et al. 2016. Assessment of Global Unconventional Oil and Gas Resources. *Pet Explor Dev* **43** (6): 925–940. [https://doi.org/10.1016/S1876-3804\(16\)30111-2](https://doi.org/10.1016/S1876-3804(16)30111-2).
- William, S. P., Kyle, J. P., David, W. S. et al. 2018. Novel Diversion Case Studies for Improved Near-Wellbore Connection between Wellbore and Hydraulic Fracture. Paper presented at the SPE/AAPG/SEG Unconventional Resources Technology Conference, Houston, Texas, USA, 23–25 July. URTEC-2881395-MS. <https://doi.org/10.15530/urtec-2018-2881395>.
- Willingham, J. D., Tan, H. C., and Norman, L. R. 1993. Perforation Friction Pressure of Fracturing Fluid Slurries. Paper presented at the Low Permeability Reservoirs Symposium, Denver, Colorado, USA, 26–28 April. SPE-25891-MS. <https://doi.org/10.2118/25891-MS>.
- Wylie, E. B. and Streeter, V. L. 1993. *Fluid Transients in Systems*. Hoboken, New Jersey, USA: Prentice-Hall.